

Chaya Silverstein

Professor Binh Dang

CSIT 166-DL DL1

28 Apr. 2026

Generic Class and Methods

In this lesson I learned how to use generics, understood how it works, and really enjoyed learning it. Thank you very much.

Explain the purpose of Generics when both `ListPrinter<String>` and `ListPrinter<Integer>` are used.

Generic data is used when you want to work with different types of data in the same class, like in the code where we used `ListPrinter<String>` to be able to work with words. When working with `<Integer>`, it works with numbers.

This is useful when I write code once using different data types in the same class. As I mentioned in my post/discussion, it can also prevent mistakes in mixing data types.

Screen shots of your output

1. There are 2 main so I deleted one - errors or bugs in your code
2. A1 A2 works
3. The result of the full code
4. The full code

My code:

```
package main;
```

```

import java.util.ArrayList;
//public
class ListPrinter<T> {
private ArrayList<T> list;
public ListPrinter(ArrayList<T> list) {
this.list = list;
}
public void printList() {
for (T item : list) {
System.out.println(item);
}
}
}

public class main {
public static void main(String[] args) {
ArrayList<String> citiesList = new ArrayList<>();
citiesList.add("San Francisco");
citiesList.add("Paris");
citiesList.add("Lima");
citiesList.add("Saigon");
citiesList.add("Bangkok");
citiesList.add("Tokyo");
citiesList.add("Cairo");
//Instantiate an object of type ListPrinter<String>
//1a. Your code goes here
ListPrinter<String>CityPrinter=new ListPrinter<>(citiesList);// i crated on object listprinter
//Invoke printList to print out citiesList
//1b. Your code goes here
System.out.println("The city list:");
CityPrinter.printList();
//2. Your code goes here
ArrayList<Integer>number=new ArrayList<>();// i create an array list
for (int i =1;i<=5;i++) { // for loop to make the list 1 to 5 :)
number.add(i);
}
ListPrinter<Integer>numberPrinter=new ListPrinter<>(number);// here I am creating anther object but for
Integer
System.out.println("the number list:");//print this
numberPrinter.printList();// print the list of numbers
}
}

```

Output:

The city list:
San Francisco

Paris
Lima
Saigon
Bangkok
Tokyo
Cairo

the number list:

1
2
3
4
5





